

## \* K-Fold Cross Validation

It is a technique where the dataset is divided into  $K$  parts, the model is trained  $K$  times using  $K-1$  parts for training and 1 part for testing, and the final performance is averaged.

For example: Total data = 100 samples

$$K=5$$

Split data into 5 parts each part 20 samples

| Run | Training data   | Testing data | Each fold becomes test data once. |
|-----|-----------------|--------------|-----------------------------------|
| 1   | Fold 2, 3, 4, 5 | Fold 1       |                                   |
| 2   | Fold 1, 3, 4, 5 | Fold 2       |                                   |
| 3   | Fold 1, 2, 4, 5 | Fold 3       |                                   |
| 4   | Fold 1, 2, 3, 5 | Fold 4       |                                   |
| 5   | Fold 1, 2, 3, 4 | Fold 5       |                                   |

Average the results

Example accuracies:

82%, 85%, 80%, 84%, 83%

Final accuracy: 82.8%